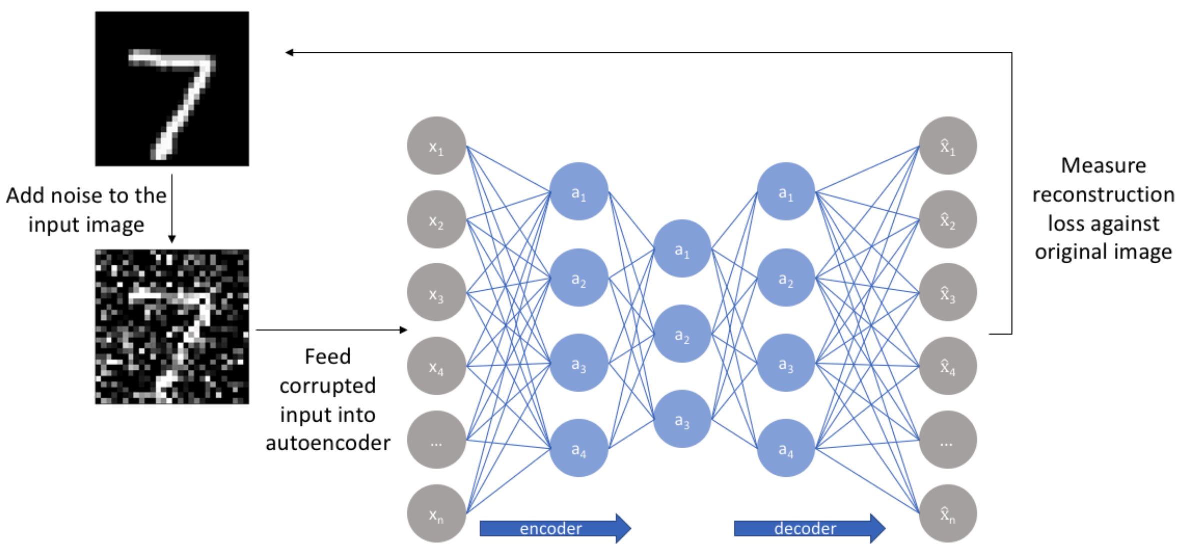
AUTOENCODERS AND VARIATIONAL AUTOENCODERS

1. Define Autoencoders, draw the architecture of the autoencoder. Execute autoencoder for a simple application of choice.

Autoencoders are a type of unsupervised neural network that learn to compress data into a compact form and then reconstruct it to closely match the original input.
It consists of an: Encoder that captures important features by reducing dimensionality.
Decoder that rebuilds the data from this compressed representation.

The model trains by minimizing reconstruction error using loss functions like Mean Squared Error or Binary Cross-Entropy. These are applied in tasks such as noise removal, error detection, and feature extraction, where capturing efficient data representations is important.

Architecture of Autoencoder



1. Encoder

It compresses the input data into a smaller, more manageable form by reducing its dimensionality while preserving important information. It has three layers which are:

- Input Layer: This is where the original data enters the network. It can be images, text features or any other structured data.
- Hidden Layers: These layers perform a series of transformations on the input data. Each hidden layer applies weights and activation functions to capture important patterns, progressively reducing the data's size and complexity.
- Output(Latent Space): The encoder outputs a compressed vector known as the latent representation or encoding. This vector captures the important features of the input data in a condensed form, which helps in filtering out noise and redundancies.

2. Bottleneck (Latent Space)

It is the smallest layer of the network that represents the most compressed version of the input data. It serves as the information bottleneck that forces the network to prioritize the most significant features. This compact representation helps the model learn the underlying structure and key patterns of the input, which helps in enabling better generalization and efficient data encoding.

3. Decoder

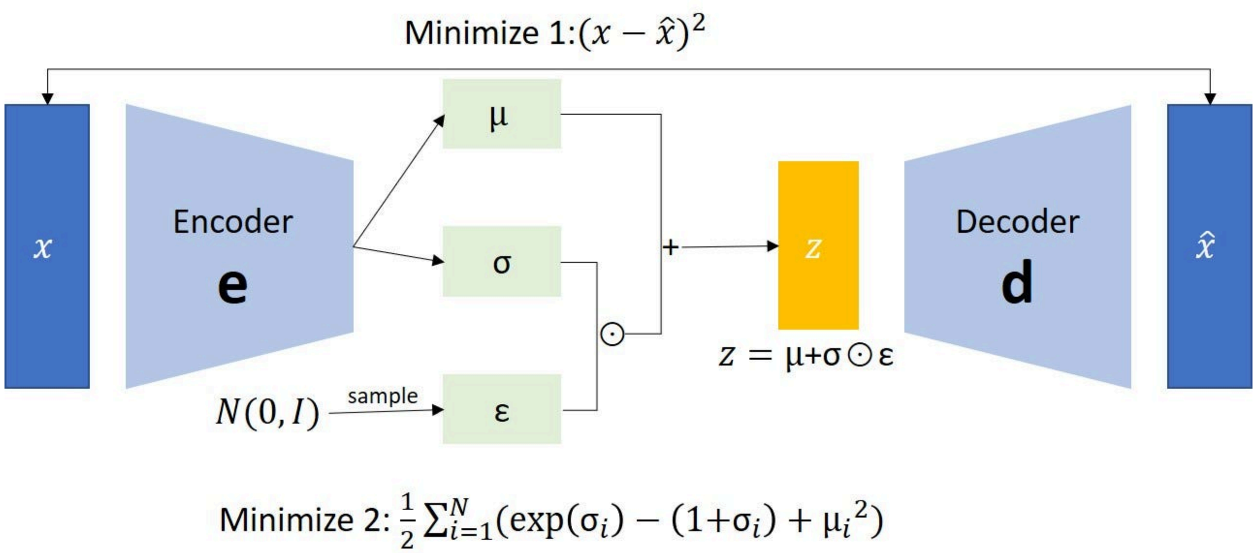
It is responsible for taking the compressed representation from the latent space and reconstructing it back into the original data form.

- Hidden Layers: These layers progressively expand the latent vector back into a higher-dimensional space. Through successive transformations decoder attempts to restore the original data shape and details
- Output Layer: The final layer produces the reconstructed output, which aims to closely resemble the original input. The quality of reconstruction depends on how well the encoder-decoder pair can minimize the difference between the input and output during training.

2. What is a Variable Autoencoder (VAE) with a sample example of your own?

Variational Autoencoders (VAEs) are generative models that learn a smooth, probabilistic latent space, allowing them not only to compress and reconstruct data but also to generate entirely new, realistic samples. VAEs capture the underlying structure of a dataset and produce outputs that closely resemble the original data.

Learns a continuous latent representation
Enables controlled and meaningful data generation
Widely used in image synthesis, anomaly detection, and representation learning



3. Compare Autoencoder and Variable Autoencoder.

Feature	Autoencoder (AE)	Variational Autoencoder (VAE)
Type of Model	Deterministic neural network	Probabilistic generative model
Latent Representation	Single fixed latent vector	Learns mean (μ) and variance (σ^2)
Latent Space	Unstructured, may have gaps	Continuous and well-structured
Sampling	Cannot sample new data	Can sample new data
Generative Ability	Limited	Strong generative capability
Loss Function	Reconstruction Loss (MSE/BCE)	Reconstruction Loss + KL Divergence
Mathematical Nature	Learns direct mapping	Learns data distribution
Output Quality	Often sharper reconstructions	Slightly blurrier reconstructions
Training Complexity	Simpler	More complex
Applications	Denoising, compression, feature extraction	Image generation, interpolation, anomaly detection